# Literature Status: Eigenvalue Convergence for Multiplicative Brownian Motion

Run `fa_banach_001` — lane 14

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**Status: literature already answered.** Ho–Zhong ask whether the empirical eigenvalue distribution of $U_N G_N(t)$ converges to the Brown measure of $ub_t$, including the case $U_N = I$. Theorem 1.2 and Definition 1.7 of Brailovskaya–Cook–Kemp–Parraud prove almost-sure weak convergence to exactly that Brown measure, in a more general initial-condition setting.

## 1 Original question

Ching-Wei Ho and Ping Zhong, *Brown Measures of Free Circular and Multiplicative Brownian Motions with Self-Adjoint and Unitary Initial Conditions*, arXiv:1908.08150, later published in *JEMS* **25** (2023), 2163–2227 [1], compute the Brown measure of $ub_t$.

On arXiv PDF page 7 they state that it remains open, even for identity initial data, to prove

$$\mu_{G_N(t)} \Longrightarrow \mu_{b_t},$$

where the left side is the empirical eigenvalue distribution and the right side is the Brown measure. On PDF page 50 they state the general unitary version: for a deterministic, or independent random, unitary $U_N$ whose empirical eigenvalue distribution tends weakly to the spectral law of a unitary $u$, prove

$$\mu_{U_N G_N(t)} \Longrightarrow \mu_{ub_t}.$$

## 2 Separate later answer

Tatiana I. Brailovskaya, Nicholas A. Cook, Todd Kemp, and Félix Parraud, *Eigenvalues of Brownian Motions on* $\mathrm{GL}(N,\mathbb{C})$, arXiv:2511.10535v2 [2], prove the following on arXiv PDF page 9 (Theorem 1.2). Let $B(t)$ be their Brownian motion on $\mathrm{GL}(N,\mathbb{C})$ and let $B_0$ be independent of it. If $B_0$ converges almost surely in $*$-distribution to $b_0$, then, for every fixed $t > 0$, the empirical eigenvalue law of $B_0 B(t)$ converges weakly almost surely to a deterministic measure $\mu_{b_0,t}$. They explicitly note that for normal $B_0$, the hypothesis is simply weak convergence of its empirical eigenvalue distribution.

Definition 1.7 on PDF page 15 identifies $\mu_{b_0,t}$ as the Brown measure of $b_0 b(t)$ and restates Theorem 1.2 in those terms. Thus $B_0 = U_N$ gives the full unitary-initial statement requested by Ho–Zhong, and $U_N = I$ gives the identity case.

## 3 Provenance and scope

This is an explicit literature closure, not a new theorem inferred by the present run. On PDF page 17, the supporting paper says that Theorem 1.2 solves the convergence problem, identifies the limiting object with the Brown measures computed in earlier work, and specifically credits Ho–Zhong as reference [68] for the unitary-initial Brown-measure computation.

The later theorem is strictly stronger in its initial-data scope: it allows arbitrary, possibly nonnormal, independent $B_0$ having an almost-sure $*$-distribution limit. No part of the source's stated identity or unitary initial-condition convergence question remains open after this theorem. A bounded local/arXiv search used the exact paper title, arXiv id, and the terms "empirical eigenvalue distribution", "multiplicative Brownian motion", and "Brown measure"; arXiv:2511.10535 was the decisive match.

## References

[1] C.-W. Ho and P. Zhong, *Brown measures of free circular and multiplicative Brownian motions with self-adjoint and unitary initial conditions*, J. Eur. Math. Soc. **25** (2023), 2163–2227; arXiv:1908.08150.

[2] T. I. Brailovskaya, N. A. Cook, T. Kemp, and F. Parraud, *Eigenvalues of Brownian Motions on* $\mathrm{GL}(N, \mathbb{C})$, arXiv:2511.10535v2 (2026).